Forward  Inverse  Example's own settings, where a change helped

**ODEs**
Integro-differential ODE †
Integro-differential ODE – Volterra kernel †
Linear ODE system †
Lotka-Volterra ODE †
Second-order ODE †
Lorenz ODE – unknown $C_1$, $C_2$, $C_3$ †
Lorenz ODE – unknown $C_1$, $C_2$, $C_3$, exogenous input †

**1D, steady**
Euler beam 1D †
Fractional Poisson 1D †
Poisson 1D – Dirichlet †
Poisson 1D – Dirichlet/Neumann †
Poisson 1D – Dirichlet/Periodic †
Poisson 1D – Dirichlet/point-set operator †
Poisson 1D – Dirichlet/Robin †
Poisson 1D – hard BCs †
Poisson 1D – multi-scale Fourier features †
Brinkman-Forchheimer 1D – unknown ν_e, K †
Diffusion-reaction 1D – unknown rate field † ‡
Fractional Poisson 1D – unknown order α †
Poisson 1D – unknown forcing field †
Poisson 1D – unknown forcing field (batches) †

**1D, time-dependent**
Allen-Cahn 1D – Wight & Zhao †
Cahn-Hilliard 1D – Wight & Zhao †
Cahn-Hilliard 1D – Mattey & Ghosh † ‡
Black-Scholes 1D – American put
Black-Scholes 1D – European call †
Burgers 1D – Raissi et al. †
Burgers 1D – RAR †
Diffusion 1D †
Diffusion 1D – hard IC and BCs †
Diffusion 1D – resampling †
Diffusion-reaction 1D †
Fractional diffusion 1D †
Heat 1D – Dirichlet †
Heat 1D – resampling †
Klein-Gordon 1D †
Schrödinger 1D †
Wave 1D – space-time multi-scale Fourier features †
Diffusion 1D – unknown coefficient †
Diffusion-reaction 1D – unknown k_f, D †

**2D, steady**
Fractional Poisson 2D – disk †
Helmholtz 2D – Dirichlet † ‡
Helmholtz 2D – hyperparameter search
Helmholtz 2D – sound-hard scattering †
Helmholtz 2D – square with hole †
Laplace 2D – disk †
Linear elasticity 2D – plate †
Linear elasticity 2D – plate, soft BCs †
Navier-Stokes 2D – Kovasznay flow †
Poisson 2D – disk †
Poisson 2D – L-shape †
Fractional Poisson 2D – unknown order α †

**2D, time-dependent**
Allen-Cahn 2D – Wight & Zhao †
Allen-Cahn 2D – Mattey & Ghosh †
Cahn-Hilliard 2D – Wight & Zhao † ‡
Cahn-Hilliard 2D – Mattey & Ghosh † ‡
Burgers 2D – Mathias et al. †
Heat 2D † ‡

**3D**
Fractional Poisson 3D – sphere †
Poisson 3D – cube with spherical cut-out †
Poisson 3D – sphere †
Allen-Cahn 3D – Wight & Zhao †
Heat 3D †
Navier-Stokes 3D – Beltrami flow †

$\omega$
$d$, $a$
$a$, $b$
C1, C3, C2
C3, C2, C1
$EI$
$K$, $\nu_e$
$\alpha$
gamma_2, gamma_1
lambda_1, lambda_2
$D$
$a$
$\gamma$
$c$
$C$
$C$
$D$, kf
$k_0$
$\mu$, $\lambda$
$Re$
$f$
$\alpha$
$\lambda\varepsilon^2$, $\lambda$
$c_2$, $c_1^2$
$\varepsilon^2$
kappa, $\alpha$
$\nu$
$\alpha$
$f$
$\lambda\varepsilon^2$, $\lambda$
$\alpha$
$\nu$

$10^{-6}$ $10^{-5}$ $10^{-4}$ $10^{-3}$ $10^{-2}$ $10^{-1}$ $10^{0}$
Relative $L_2$ error of the main output at the last reference time

0.001 0.1 10
Worst parameter error (%)

† in this study (filled: the chosen run); rows without † show the README's result. ‡ the run with the lowest training loss (shown) is more than 2× less accurate than another run.